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Mergers, Product Prices, and Innovation: Evidence from the Pharmaceutical Industry

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ABSTRACT

Using novel data from the pharmaceutical industry, we study product prices and innovation around mergers. Exploiting within-deal variation in product market consolidation, we show that prices increase more for drugs in consolidating markets than for matched control drugs. Estimates indicate a 2% average price effect that persists for about one year. Price increases expand with acquirer-target product similarity and are more pronounced within less competitive product markets with fewer players and no generic competition. Examination of trade-offs reveals these deals generate significant shareholder value. They also spur labeling and other manufacturing-related innovation, but not the development of new drugs.

THAT MERGERS AND ACQUISITIONS GENERATE shareholder value is generally accepted (Betton, Eckbo, and Thorburn (2008)). How mergers benefit shareholders and how they affect the costs that other stakeholders incur are less clear. Industry consolidation can create value through synergies, leading to efficiency gains and lower product prices (Sheen (2014)). Mergers can also result in technology sharing, which stimulates innovation (Bena and Li (2014)). Alternatively, some mergers may concentrate market power and suppress competition (Cunningham, Ederer, and Ma (2021)), leading prices to drift upwards and innovation to decline.

This study examines changes in product prices and innovation around consolidation in the pharmaceutical industry. The pharmaceutical industry represents an ideal laboratory for testing the effects of mergers on product markets.

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To begin, comprehensive and reliable price data are available at the product, as opposed to the firm or industry, level. Another advantage of this industry is its standardized product classification systems, which allow us to pinpoint shifts in product market competition around mergers that join producers of competing products. Finally, quality often changes around mergers in other product markets, conflating the effects of mergers on price, while in the pharmaceutical industry, we can assume that quality is held constant because product codes correspond to precise doses of specified ingredients. These unique industry features thus enable more precise identification of changes in product market outcomes around mergers than in previously studied industries.

In addition, understanding the drivers of prices and innovation in the pharmaceutical industry is valuable in its own right because its products directly contribute to social welfare. For this reason, policymakers make access to pharmaceutical products a top priority. For example, rising drug prices in the United States prompted the February 2019 hearing during which pharmaceutical executives testified before the Senate Finance Committee as well as the July 2021 Senate Antitrust Subcommittee hearing with industry experts and patient advocates during which Republican Senator Mike Lee of Utah argued that "innovation, competition, and affordable prices are arguably more important to prescription drug markets than almost any other consumer market we can think of." Determining whether consolidation in this industry tends to intensify or mitigate rising prices and new drug development therefore informs policy debates at the highest level of politics.

Are pharmaceutical mergers ultimately associated with cost efficiencies, or are they instead associated with price increases due to the concentration of market power? To test this question, we obtain detailed drug price data from the National Average Drug Acquisition Cost (NADAC) survey conducted for the Centers for Medicare and Medicaid Services (CMS), which restricts our sample to the 2013 to 2019 period. NADAC survey drug prices represent the average unit cost of drugs to retail pharmacies, the customers of pharmaceutical manufacturers. When we combine these price data with merger and acquisition (M&A) announcements from the Securities Data Corporation, we identify 202 pharmaceutical deals, worth an aggregate $687 billion, in which the acquirer produces drugs with available price data.

Our identification strategy hinges on exploiting within-deal variation in expected product market consolidation. We construct a difference-in-differences (DID) setting comparing price changes of acquirer-firm drugs whose product markets consolidate with simultaneous price changes of matched, same-firm control drugs whose product markets are not directly affected by the merger. Consolidation should occur when the acquirer and the target share a product market. We define product markets using Anatomical Therapeutic Chemical (ATC) codes from the World Health Organization (WHO). To control for drug features that affect prices or consolidation likelihood, we exactly match sampled drugs to control drugs on prescription versus over-the-counter and brand name versus generic statuses using NADAC data and on patent and exclusivity rights coverage using data from the U.S. Food and Drug Administration

(FDA) "Orange Book." We then select the matched control product with the closest lagged reimbursement volume, a proxy for demand, from Medicaid's State Drug Utilization Database (SDUD). We confirm there are no significant differences in premerger price trends across sample and control products, consistent with the parallel trends assumption being satisfied.

We then construct our DID regression using the natural logarithm of drug prices as the dependent variable. Our coefficient of interest corresponds to the interaction between an indicator variable for drugs with product market consolidation and an indicator corresponding to postmerger time periods. It measures the difference in (approximate percentage) price changes around mergers between drugs whose product markets consolidate and matched control drugs. Because our DID sample is constructed at the deal level, we "saturate" product and time fixed effects with deal indicators. Product-deal fixed effects net out differences in price levels across products at the time of the merger, and deal-event-time fixed effects account for general price trends around the merger.

Relative to matched control drugs, acquirer drugs with expected product market consolidation experience significantly greater price increases. Their prices increase 2.2% more and remain elevated for approximately one year. Closer analysis of the price dynamics reveals that price trends do not differ leading up to the merger (consistent with parallel trends) but diverge as early as the quarter of the merger.[1] Further, this price differential is not due to control drug prices decreasing; inflation-adjusted prices increase around mergers for both sample and control drugs, but more so for sampled drugs. Overall, our findings show that acquiring firms raise prices more within consolidating product markets. These results are inconsistent with synergy gains within shared product markets being passed along to consumers.

Our findings withstand a battery of robustness tests. We first verify the robustness of our results to alternative methods of identifying acquirer/target product overlap. We exploit the nested, hierarchical nature of the ATC code system to narrow our definition of acquirer/target overlap. Our baseline overlap definition uses therapeutic subgroups from three-digit or "2nd level" ATC codes, but we also redefine overlap using codes for pharmacological subgroups ("3rd level"), chemical subgroups ("4th level"), and the chemical substance ("5th level"). Not only do our main findings survive alternative overlap definitions, but we also observe that price differences expand with product similarity. For instance, when we narrow our overlap threshold to the highest ATC code level, our coefficient of interest almost doubles, implying incremental price increases above 4%. If market power is the channel through which mergers relate to

[1] Quick price responses are consistent with anecdotal evidence suggesting that price changes around mergers can be immediate. For example, Celgene increased the price of its top-selling drug Revlimid 3.5% on the day its planned deal with Bristol-Myers Squibb was announced. ("Pharmaceutical Industry CEOs Face Senate Hearing on Drug Prices," *The Wall Street Journal*, February 25, 2019.) Another example involves Ovation Pharmaceuticals' Indocin IV price hike from $109 per treatment to $1,500 per treatment two days after purchasing competing drug NeoProfen from Abbott Laboratories. (Klobuchar, Amy, 2020, *Antitrust: Taking on Monopoly Power from the Gilded Age to the Digital Age* (Alfred A. Knopf, New York)).

prices, then we expect price increases to rise with product similarity, as we find; these findings provide us with some confidence that our results are a function of the market power channel and are not spurious. We further examine our estimates' sensitivity to constructing product markets based on a drug's mechanism of action (MoA) or established pharmacological class (EPC). Our results continue to hold. We also verify robustness to "cleaner" subsamples of drugs by eliminating drugs involved in multiple mergers in close proximity. We obtain estimates ranging from 1.6% to 3.6%, similar to those in our baseline regressions. Finally, while we focus primarily on drugs produced by the bidding firm in a merger, we confirm that drugs manufactured by target firms also tend to experience price increases of comparable magnitudes.

We end our price analysis by investigating cross-sectional heterogeneity in price changes around acquisitions using a triple-differences model. If market power drives the relation between mergers and drug prices, then price increases should be greater in less competitive product markets, where firms could more easily exert market power due to lower coordination costs or greater expected gains in market share. We indeed find that prices increase more in markets with fewer participants and with no generic competitors, suggesting that consolidation within less competitive markets is more detrimental to consumers. In fact, we observe no price effect of mergers in disperse product markets. These findings are important because they identify the types of product markets—those lacking competition, particularly from generics—most susceptible to price hikes around mergers.

We conclude by quantifying two potential benefits to pharmaceutical mergers. The first involves shareholder value creation. On average, pharmaceutical merger announcements are associated with positive and significant five-day acquirer cumulative abnormal returns (CARs) of 1.5%, consistent with substantial shareholder value creation. Shareholder gains are greater around deals that consolidate market power the most, that is, horizontal mergers and deals with greater or more economically meaningful acquirer/target product market overlap. If mergers generate shareholder wealth through synergistic cost-cutting (e.g., eliminating redundancies in administration or distribution), then we would not expect merger announcement returns to vary with proxies for market power consolidation, as we find. Our findings instead lend support to the view that mergers benefit shareholders more when they increase market power.

The second trade-off we examine is innovation. We depart from traditional innovation measures based on patents because pharmaceutical patents are subject to concerns of patent "thicketing," that is, applying for multiple patents on the same drug to block competition.[2] We thus create alternative innovation proxies based on FDA New Drug Approval data. Unlike prices, innovation metrics are measured at the firm level. We adjust our methodology accordingly.

[2] Pharmaceutical industry experts claim that patents fail to represent "true innovation." ("Pharma patent owners in the U.S. are under pressure like they have never been before," *IAM*, November 26, 2018.)

Our innovation DID model is analogous to our price model, except that it compares innovation changes within acquiring firms to innovation changes within matched nonacquiring firms. To control for selection into acquisition activity, we pair each acquiring firm with the nonacquiring firm closest in propensity to acquire, estimated using the Harford (1999) acquisition likelihood model. We run regressions at the deal level, including deal-firm and deal-event-time fixed effects.

We find some evidence that new drug applications increase more within acquiring firms than control firms. This increase, however, is driven by follow-on applications, the majority of which relate to changing an existing product's label or manufacturing process. Initial new drug applications—which arguably encompass the most innovative applications, such as applications for approval of new molecules and new active ingredients—do not increase around mergers. We obtain similar findings when we condition on acquisitions of biotech companies, often considered to be the most innovative pharmaceutical firms, and when we extend our sample to examine new drug applications up to 10 years postmerger to account for new drug development being a lengthy process.

This study contributes to the ongoing debate about the winners and losers in mergers and acquisitions. Many other studies focus on specific industries and document detrimental consumer outcomes around mergers, consistent with our findings. Airfares increase more on routes served by merging airlines relative to routes unaffected by mergers (Kim and Singal (1993), Kwoka and Shumilkina (2010)). Hospital mergers are associated with increases in prices but not quality of care (Cooper, et al. (2019), Dafny (2009), Vita and Sacher (2001)), and rival hospitals increase prices around mergers as well (Dafny (2009)). Eliason, et al. (2020) show that recent consolidation in the dialysis industry is associated with increased reimbursements but worse patient outcomes, consistent with a decline in value. Deposit rates decline following bank mergers, though these adverse price effects are temporary (Prager and Hannan (1998), Focarelli and Panetta (2003), Garmaise and Moskowitz (2006)). Around mergers, petroleum companies increase oil prices (Hosken, Silvia, and Taylor (2011)), especially wholesale prices (Taylor and Hosken (2007)). Mergers are also associated with price hikes of everything from academic journals (McCabe (2002)) and household appliances (Ashenfelter, Hosken, and Weinberg (2013)) to beer (Miller and Weinberg (2017)). In contrast, several studies find null effects or product price declines around mergers. Sapienza (2002) documents that interest rates charged by banks decrease after consolidation, but, as market shares increase, these efficiency gains disappear and credit supply to small borrowers declines. Reexamining the price effects of airline mergers, Luo (2014) finds no significant price changes after the merger of Delta Airlines and Northwest Airlines. Using *Consumer Reports* data, Sheen (2014) finds that, when two firms selling common products merge, the quality of the related products converges and the price drops, although these efficiency gains take two to three years to be realized. In sum, whether market power or efficiency gains dominate in mergers is an unresolved question and likely depends on the industry and product market under consideration.

We extend the above literature in several unique and important ways. First, unlike many seminal studies in this area that focus on a single merger, we study an entire industry. This allows us to obtain a larger, more representative merger sample and to examine cross-sectional variation at the deal level in addition to the product level. Second, relative to other industries, the pharmaceutical industry provides important empirical advantages—granular product-level price data, products with relatively time-invariant quality, a widely accepted product grouping system, and unique innovation milestones—that allow us to contribute to the literature on mergers and market power by more cleanly identifying shifts in product market competition and simultaneous changes in product prices, innovation, and shareholder value. Last but not least, given the substantial welfare benefits of pharmaceutical products, understanding their price and innovation patterns is a valuable exercise. Our findings suggest that even within this large and important industry at the center of national policy debates, mergers are associated with significant price increases. A back-of-the-envelope estimate suggests that mergers contribute to $1.5 billion in extra U.S. government spending per year on prescription drugs alone.[3] Importantly, we also show that pharmaceutical mergers do not appear to foster groundbreaking innovation, a common justification for high drug prices.[4] Our findings therefore inform policy around drug pricing by proposing careful antitrust enforcement as one potential remedy for rising drug prices. Even so, the welfare costs from higher drug prices associated with these mergers should be weighed against their benefits to the owners of the firm as well as the costs associated with increasing enforcement.

Our paper is closely related to, but nonetheless distinct from, several recent studies on pharmaceutical mergers. Haucap, Rasch, and Stiebale (2019) and Cunningham, Ederer, and Ma (2021) both document declines in innovation around pharmaceutical mergers. Studying large pharmaceutical mergers in Europe, Haucap, Rasch, and Stiebale (2019) show a decrease in patent applications by merging firms and their rivals, albeit to a lesser extent. Discrepancies in our results likely stem from different merger samples and time periods as well as different innovation metrics. While Haucap, Rasch, and Stiebale (2019) do not include acquisitions of small biotech firms, Cunningham, Ederer, and Ma (2021) explicitly focus on drugs in the early stages of development. They show evidence of firms acquiring competing drug manufacturers to discontinue development and thus squelch competition. These so-called "killer acquisitions" constitute between 5.3% and 7.4% of pharmaceutical mergers,

[3] In 2019, prescription drug spending for Medicare Part B, Medicare Part D, and Medicaid reached $37 billion, $183 billion, and $69 billion, respectively, for a total of $289 billion. In our sample, drugs involved in a merger impacting their product market account for of 23.1% reimbursement dollars, and these drugs are associated with price increases of 2.2%. This translates into $1.5 billion ($0.231 \times 0.022 \times$ $289 billion) in extra U.S. government spending per year.

[4] For example, during his testimony before the Senate Finance Committee in February 2019, Sanofi's CEO Olivier Brandicourt referenced new medications and research and development expenses (https://www.finance.senate.gov/imo/media/doc/26FEB2019BRANDICOURT-SANOFI.pdf).

according to their most conservative estimates. Our study complements theirs as we instead focus mainly on drug *prices*. In a subsequent working paper, Hammoudeh and Nain (2020) also study pharmaceutical mergers. Although their univariate tests are consistent with price increases around mergers, as we find, parts of their multivariate analysis suggest reductions in price. The most likely explanations for these results are differences in data sources and empirical strategies.[5] Other unique features of our study include our examinations of changes in new drug applications and shareholder value around mergers.

The remainder of the paper is organized as follows. Section I formalizes our hypotheses. Section II describes our data and sample construction. Section III motivates our identification strategy and presents our main results, robustness checks, and cross-sectional analyses. Sections IV and V examine shareholder value creation and innovation around mergers. Section VI concludes.

I. Hypothesis Development

The M&A literature documents evidence of mergers creating shareholder value (Betton, Eckbo, and Thorburn (2008)). The two primary sources of this value creation are efficiency gains (i.e., "synergies") and enhanced market power.

Several studies find support for postacquisition efficiency gains by providing evidence of improved operating performance, elimination of redundancies, and increased innovation. For example, Healy, Palepu, and Ruback (1992) and Heron and Lie (2002) show that merged firms exhibit superior postmerger operating performance relative to industry peers, and Hoberg and Phillips (2010) show that mergers can improve profit margins through product differentiation. Maksimovic, Phillips, and Prabhala (2011) document that manufacturing firms sell 27% and close 19% of target firms' plants after mergers. Lee, Mauer, and Xu (2018) document better postmerger performance when merged firms have related human capital due to reduced labor costs. Bena and Li (2014) further show that premerger technology overlap has a positive effect on future innovation, consistent with mergers improving innovation capabilities through synergies. These studies suggest that synergies are an important source of value creation around mergers and therefore motivate our first hypothesis, the *Synergistic Gains Hypothesis*. Under this hypothesis, to the extent that mergers can lead to synergistic gains through the elimination of redundancies and the sharing of technology, firms may pass along gains to consumers in the

[5] Hammoudeh and Nain (2020) estimate prices using Medicaid reimbursements from states to pharmacies. States historically overpaid for reimbursements, which prompted the CMS to collect price estimates using the NADAC survey, our data source. Furthermore, Hammoudeh and Nain (2020) infer acquirer/target product market overlap using textual analysis to calculate similarity scores between descriptions of therapeutic area and MoA, whereas we directly match drugs using ATC codes and MoA or EPC for robustness. Finally, we match affected drugs to control drugs within the same firms with identical characteristics, while their controls come from nonacquiring firms.

form of lower prices. The *Synergistic Gains Hypothesis* leads to three empirical predictions. First, it predicts that product prices decline around mergers. Second, because redundancies are more likely to occur within similar product markets, this hypothesis predicts that price declines should be greatest within product markets shared across the bidder and target. Third, this hypothesis also predicts that technology sharing should lead to more innovation.

Alternatively, mergers can strengthen market power, which may harm customers and suppliers. For example, using the industry-level Producer Price Index from the Bureau of Labor Statistics, Fathollahi, Harford, and Klasa (2022) document price increases around horizontal mergers, particularly within industries with high product similarity, and Bhattacharyya and Nain (2011) show that mergers allow firms to exert price pressure on suppliers. Accordingly, our second hypothesis, the *Market Power Hypothesis*, posits that because mergers consolidate market power, they may adversely impact stakeholders. The broad prediction of the *Market Power Hypothesis* is that merging firms exploit enhanced market power by charging higher prices to customers. Price changes should be greatest in product markets in which market power is most consolidated, specifically, product markets that the acquirer and target have in common.

Because synergies and enhanced market power are not mutually exclusive, our empirical estimates capture the *net* effect of mergers on customers and therefore reveal which force dominates.

II. Data Sources, Sample Construction, and Summary Statistics

This section describes the construction of our samples of drug prices, other drug features, innovation measures, and M&A deals. We also summarize and discuss our main variables.

A. Drug Price Data and Trends

Our drug price proxy is based on data from the NADAC survey conducted for the CMS. Prior to the NADAC survey, the CMS allowed states to independently estimate drug costs for Medicaid reimbursements. Most states based cost estimates on average wholesale prices (AWP). In 2011, the Office of Inspector General reported that AWP-based reimbursement rates were "fundamentally flawed," leading to inflated estimates, and recommended a single national benchmark that more accurately reflects actual pharmacy costs and thereby "eliminate[s] States' reliance on the inflated published prices that cause Medicaid and its beneficiaries to pay too much for certain drugs."[6] To develop a credible pricing benchmark, in 2012, the CMS began contracting with Myers and Stauffer, a national certified accounting firm, to survey retail pharmacies. Myers and Stauffer collects acquisition costs for covered outpatient drugs from a random sample of retail pharmacies designed to closely align with the

[6] See https://oig.hhs.gov/oei/reports/oei-03-11-00060.pdf.

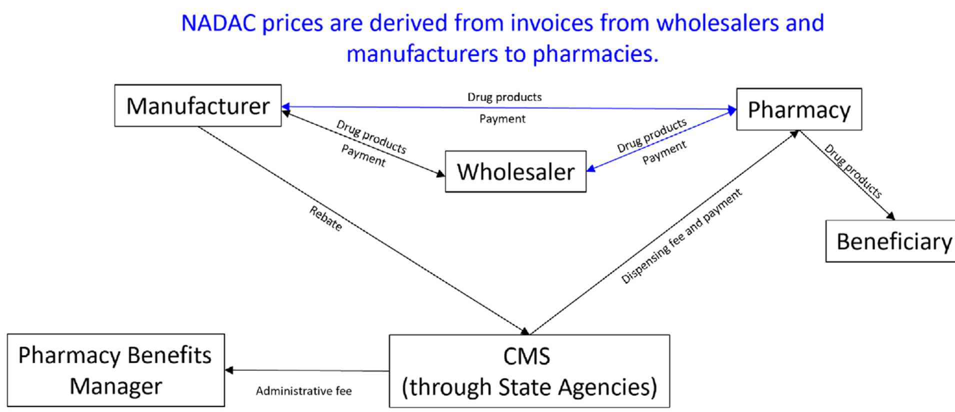

Figure 1. Pharmaceutical industry structure and main players. This figure describes the payment and supply chain for prescription drug benefits through the Center for Medicaid Services (CMS). National Average Drug Acquisition Cost (NADAC) prices are derived from invoices from manufacturers and wholesalers to pharmacies, denoted by the blue lines. NADAC prices do not reflect off-invoice discounts, rebates, or price concessions. Sources: https://www.kff.org/medicaid/issue-brief/pricing-and-payment-for-medicaid-prescription-drugs/ and https://www.medicaid.gov/medicaid/prescription-drugs/downloads/retail-price-survey/nadac-overview-operations.pdf. (Color figure can be viewed at wileyonlinelibrary.com)

composition of the population of U.S. pharmacies, that is, representing all 50 states and the District of Columbia and including both independent and chain pharmacies.[7] As shown in Figure 1, NADAC prices are derived from invoices from pharmaceutical manufacturers and wholesalers to pharmacies and do not reflect off-invoice discounts or rebates.

The NADAC therefore represents the average cost that retail pharmacies pay for drugs. This measure of drug prices has advantages and disadvantages. The NADAC is ideal for testing our research question—how mergers of pharmaceutical manufacturers influence customers—because retail pharmacies are drug makers' customers. The NADAC is less appropriate for precisely quantifying the price that consumers ultimately pay, which depends on retail pharmacies' profit margins as well as patients' insurance coverage. Even so, it seems plausible that pharmacies will pass along at least a portion of price changes to end users. We also note that CMS reimbursement rates are based on NADAC prices, and all taxpayers bear the cost of government reimbursements.

The NADAC survey results allow for calculation of retail pharmacies' average acquisition cost per unit at the National Drug Code (NDC) level. Visual inspection of the data suggests that "units" generally represent one individual tablet or capsule, tube of cream/gel/ointment, bottle of spray, or vial of

[7] A more detailed description of the CMS's methodology can be found at https://www.medicaid.gov/medicaid-chip-program-information/by-topics/prescription-drugs/ful-nadac-downloads/nadacmethodology.pdf.

NDC Description	NADAC Per Unit ($)	Effective Date	Days Effective in 2018Q3	Weight (Days Eff./ Total days)	NADAC * Weight
LIPITOR 40 MG TABLET	14.83	7/1/2018	15	0.163	2.4187
LIPITOR 40 MG TABLET	13.55	7/16/2018	37	0.402	5.4515
LIPITOR 40 MG TABLET	13.65	8/22/2018	40	0.435	5.9352
Weighted average quarterly price:					13.81

Figure 2. Quarterly drug price measure: An example. This figure illustrates our method of calculating quarterly drug prices using Lipitor as an example. Using drug prices and effective dates from the National Average Drug Acquisition Cost (NADAC) survey conducted for the Centers for Medicare and Medicaid Services (CMS), we create a time-weighted average drug price each quarter (top). Our weightings are based on the number of days in which the price was effective during the quarter, scaled by the total number of days in the quarter. We provide an example of the calculation of Lipitor's 2018Q3 price (bottom). (Color figure can be viewed at wileyonlinelibrary.com)

medicine, ordinarily associated with a specified dose. We are careful to ensure that quantities remain constant over time.

Because the NADAC survey reports unit price data with irregular frequencies, we aggregate prices to the quarterly level by weighting the NADAC by the number of days between reported price changes. Figure 2 presents an example of our quarterly price measure using 2018Q3 NADAC prices for the 40 mg Lipitor tablet. First, we calculate the number of days between effective dates associated with each NADAC price and divide by the number of days in the quarter to obtain a time weighting for each price. For example, a new price ($14.83) is posted at the beginning of the quarter on July 1, 2018. We assume that this price is valid for 15 days until the next price ($13.55) becomes effective on July 16, 2018. This second price is then effective for 37 days until August 22, 2018, when the final price ($13.65) is posted. The corresponding weights are thus

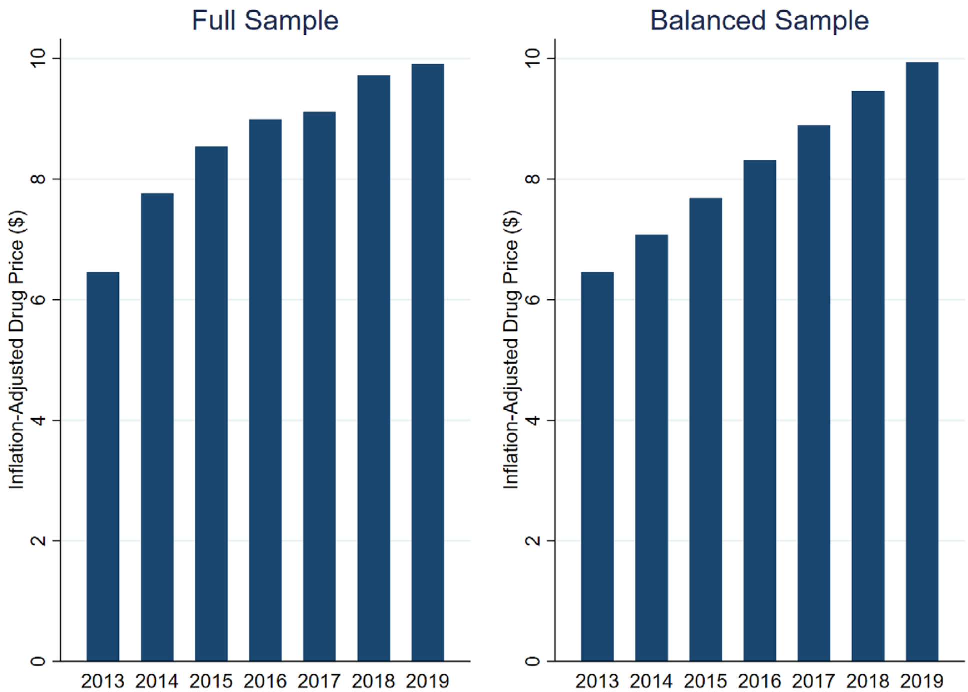

Figure 3. Drug price trends. This figure presents the average year-end (fourth quarter) unit price (in 2019 U.S. dollars) of drugs sold to retail pharmacies for our full sample of drug prices (left) and for a balanced sample requiring available price data each year (right). We source drug prices from the National Average Drug Acquisition Cost (NADAC) survey conducted for the Centers for Medicare and Medicaid Services (CMS). (Color figure can be viewed at wileyonlinelibrary.com)

0.163 (15/92), 0.402 (37/92), and 0.435 (40/92). We multiply each price by its weight and sum the weighted prices to obtain the time-weighted average quarterly price for each drug product. Finally, we adjust for inflation using the Consumer Price Index (CPI) and express all prices in 2019 dollars. The availability of drug prices from the NADAC survey restricts our drug price sample period to 2013 to 2019. The full NADAC sample consists of 805,155 drug-quarter observations between 2013 and 2019 associated with 41,395 unique NDCs.

Figure 3 shows that the average annual inflation-adjusted drug price per unit follows a strong, upward trend. After inflation adjustments, drug prices within the full sample average $6.46 per unit in 2013 but $9.91 per unit by 2019. To ensure that observed price increases are not driven by market entry and exit, we examine price trends within a balanced sample of drugs with price data available all seven years. We find that prices increase monotonically within the balanced sample as well. For the balanced sample, inflation-adjusted unit drug prices average $6.46 in 2013 but grow to $9.94 by 2019.

Table I, Panel A, reports summary statistics on drug prices and characteristics for the full NADAC sample. The mean price per unit is approximately $9.16, and the median is $0.24. The distribution is right-skewed, indicative of

Table I

Summary Statistics

This table presents summary statistics on drugs characteristics, pharmaceutical innovation metrics, and pharmaceutical M&A announcements between 2013 and 2019. Panel A presents statistics at the drug-quarter level for the full sample of drugs covered in the Centers for Medicare and Medicaid Services (CMS) survey. *Price* is the National Average Drug Acquisition Cost (NADAC) per unit reported by surveyed retail pharmacies, averaged over the fiscal quarter of the drug manufacturer and expressed in 2019 dollars. Created from the NADAC database, *Brand name*, *Generic*, *Prescription*, and *Over-the-counter* are indicator variables for these drug characteristics. *Patent* (*Exclusivity*) is an indicator equal to one if the drug is under patent (exclusivity protected), as noted in the Food and Drug Administration (FDA) database. *Units reimbursed* are the total units reimbursed, expressed in millions, in the prior quarter as reported in Medicaid's State Drug Utilization Data (SDUD). Panel B presents innovation metrics derived from the FDA New Drug Approval database. *All applications* represent all FDA new drug applications, aggregated at the firm-quarter level. First, we segment new drug applications into *Initial* or "original," applications and *Secondary* or "supplemental" applications, representing changes to an FDA-approved application. We then form three groups of applications by submission classification code, illustrated in Internet Appendix Table IA.I: *New drug*, *Labeling*, and *Other & missing*. Panel C presents summary statistics for M&A announcements in the Securities Data Corporation (SDC) database between 2013 and 2019 associated with drug prices reported in the NADAC survey. *Deal value* represents the dollar value of the transaction, in billions. *Public target*, *Private target*, and *Subsidiary* are indicators denoting whether the target firm is publicly traded, privately held, or a subsidiary of the acquiring firm, as reported in SDC.

	Mean	Std. Dev.	P10	Median	P90
Panel A: Drug characteristics					
Price	9.163	179.900	0.027	0.244	6.586
Brand name	0.140	0.347	0	0	1
Generic	0.860	0.347	0	1	1
Prescription	0.855	0.353	0	1	1
Over-the-counter	0.145	0.353	0	0	1
Patent	0.071	0.257	0	0	0
Exclusivity	0.020	0.139	0	0	0
Units reimbursed (millions)	0.302	2.767	0	0.004	0.445
Panel B: Innovation metrics					
All applications	33.454	62.264	0	6	90
Initial	3.646	9.525	0	0	11
Secondary	29.808	56.005	0	5	85
New drug	0.536	1.080	0	0	2
Labeling	19.620	42.375	0	2	56
Other & missing	13.297	23.164	0	3	36
Panel C: Deal characteristics					
Deal value (in $billions)	3.400	9.115	0.095	0.670	8.662
Public target	0.297	0.458	0	0	1
Private target	0.347	0.477	0	0	1
Subsidiary	0.356	0.480	0	0	1

a small percentage of drugs in our sample being quite expensive. For example, while the 90th percentile drug price is \$6.59 per unit, the 99th percentile hovers around \$96.57 per unit.[8,9] The NADAC survey also distinguishes between brand name versus generic, and prescription versus over-the-counter, drugs: Within the full NADAC sample of drugs, 14% are brand name (as opposed to generic) and 86% are prescription (as opposed to over-the-counter). We obtain additional drug characteristics from several other sources. We collect patent and exclusivity rights from the FDA's Approved Drug Products with Therapeutic Equivalence Evaluations database, commonly known as the Orange Book. We match patent and exclusivity data with the NDC Directory by new drug application type and number, and then merge to our drug acquisition cost data using NDC. We classify a drug as patented or covered under exclusivity rights if its patent or exclusivity has not yet expired. We find that 7% of drugs in the full NADAC sample are under patent, and 2% are exclusivity protected.[10] Ideally, we would like to control for demand with drug-level sales volume. But drug manufacturers are not required to report sales at the product level. We instead turn to Medicaid's SDUD, which reports total units and amounts reimbursed by NDC by state. We aggregate these state-level figures to reflect the total units of each product reimbursed by Medicaid each quarter. If Medicaid reimbursement units are correlated with total units sold, our reimbursement metric should serve as a proxy for demand. Medicaid reimburses 302,000 units of the average drug product in our sample, but this metric is highly skewed: The median reimbursement represents only 4,000 units, while the 90th percentile is nearly half a million units.

B. Innovation Metrics

Although studies of other industries generally rely on patent volume and citations to measure innovation, these metrics are problematic within the pharmaceutical industry. Pharmaceutical companies have been accused of patent thicketing, a competition-blocking practice of applying for multiple patents on the same drug or extending patents on existing drugs. For instance, AbbVie has obtained over 100 patents on Humira, the world's best-selling drug,

[8] Our findings are not driven by extreme outliers such as the 2015 scandal involving "Pharma Bro" Martin Shkreli, whose company Turing Pharmaceuticals acquired the drug Daraprim, which treats a life-threatening parasitic infection, and increased its price over 5,000% ("Drug Goes From \$13.50 a Tablet to \$750, Overnight," *The New York Times*, September 20, 2015). Our results are similar in magnitude and significance when we winsorize prices at the 1st and 99th percentiles. See Internet Appendix Table IA.VI.

[9] The Internet Appendix is available in the online version of the article on *The Journal of Finance* website.

[10] Patents are broader property rights issued any time during the development period, while exclusivity rights are issued only upon drug approval and refer to specific delays and prohibitions on rival drug approval. Also, patents generally span 20 years, although they can be extended, whereas exclusivity rights last 180 days to seven years, depending on the type of drug. For more details, see https://www.fda.gov/drugs/development-approval-process-drugs/frequently-asked-questions-patents-and-exclusivity.

alone.[11] This practice is not uncommon. On average, the United States' 12 best-selling drugs hold 71 patents per drug, with exclusivity lasting 38 years, almost twice the standard 20-year exclusivity period for core patents.[12] According to Rachel Sher, the deputy general counsel of the Association for Accessible Medicines, which represents generic drug companies, "Too often branded companies are seeking to patent features of the drugs that don't represent true innovation."[13]

To address the aforementioned patent thicketing problem, we use new drug approval data from the FDA to create a cleaner innovation proxy. The FDA reports all approved drug applications and categorizes them as initial versus secondary applications, as well as by submission classification. Initial, or "original," applications represent first-time applications whereas secondary, or "supplemental," applications request a change to an FDA-approved application. Submission classifications (illustrated in Table IA.I) broadly cover the creation of new drugs (e.g., new molecules, new active ingredients, or new combinations), labeling changes, and other applications, with the most common "other" category being a process optimization known as "chemistry, manufacturing, and control" or "CMC." All new drug applications are contained within initial applications, and all labeling applications are considered secondary applications. Other and missing submission codes appear in both initial and secondary applications.

We aggregate drug applications (total and by category) at the firm-year level and present summary statistics in Table I, Panel B. On average, drugs manufacturers submit 33.5 applications per year, of which 3.6 are first-time applications and 29.8 are follow-on applications for existing products. The average firm submits only 0.5 applications per year corresponding to new drugs. The bulk of applications instead relate to labeling (19.6 per firm-year on average) or manufacturing and other process changes (13.3 per firm-year). Arguably, initial applications are more innovative than supplemental applications, and new drug applications are more innovative than labeling or manufacturing changes. To the extent this is true, actual innovation is rare: Only 10% of all applications are initial applications, of which only 1.6% correspond to new drug products.

C. Pharmaceutical Mergers

We obtain M&A announcements from Securities Data Company (SDC) Platinum. Following prior literature (e.g., Bena and Li (2014)), we identify M&As

[11] "By Adding Patents, Drugmaker Keeps Cheaper Humira Copies Out of U.S." *The Wall Street Journal*, October 16, 2018.

[12] "Overpatented, Overpriced: How Excessive Pharmaceutical Patenting is Extending Monopolies and Driving up Drug Prices," *Initiative for Medicines, Access & Knowledge (I-MAK)*, August 2018.

[13] "Pharma patent owners in the U.S. are under pressure like they have never been before," *IAM*, November 26, 2018.

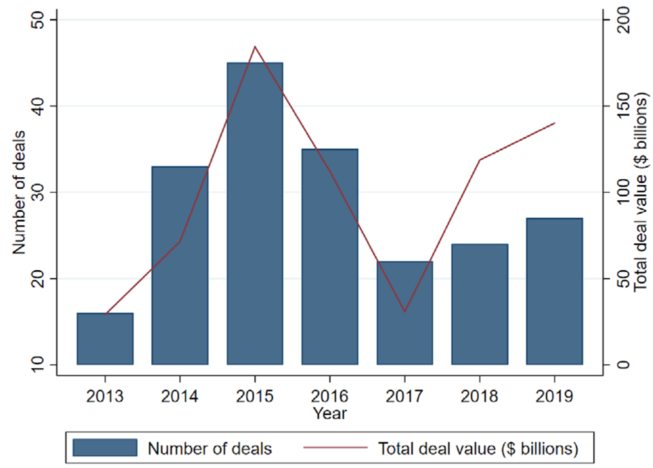

Figure 4. Pharmaceutical mergers and acquisitions. This figure presents the number (left axis) and total deal value in billions of U.S. dollars (right axis) of pharmaceutical mergers and acquisitions (M&As) each year. Our merger sample consists of firms with M&A announcements in the Securities Data Corporation (SDC) database between 2013 and 2019 and with drug prices reported in the National Average Drug Acquisition Cost (NADAC) survey conducted for the Centers for Medicare and Medicaid Services (CMS). (Color figure can be viewed at wileyonlinelibrary.com)

using deal codes corresponding to a merger, an acquisition of majority interest, or an acquisition of assets. We condition on completed deals in which the acquirer owned less than 50% of the target firm prior to the deal and at least 90% after the deal. To merge our drug price data with SDC, we first identify drug manufacturers (i.e., "labelers") using the NDC Directory, which provides information submitted to the FDA on labeler names for each product. If necessary, we identify the parent company associated with the labeler through web searches. For example, Johnson & Johnson is the parent company of the labeler Janssen Pharmaceuticals. We use a fuzzy name match to link drug manufacturers directly to acquirer names in SDC or indirectly by first matching with company names in the Center for Research in Security Prices (CRSP)/Compustat merged database, and then merging with SDC on CUSIP. The CRSP/Compustat merged database also serves as the source of our firm-level accounting and stock price data.

Figure 4 gives an overview of M&A activity within the pharmaceutical industry from 2013 to 2019. Deal volume peaks at 45 deals in 2015, decreases to 22 deals in 2017, and rebounds some in 2018 and 2019. Deal values follow a similar pattern: Aggregate deal value spikes in 2015 at $184 billion and bottoms out in 2017 at $31 billion. The aggregate value of the 202 M&As in our sample is $687 billion. These deals are conducted by 67 unique acquirers. Table I, Panel C, shows that these deals are substantial in size: The average deal is worth $3.4 billion, and even deals at the 10^{th} percentile are valued at $95 million. Acquisitions of publicly traded targets, private targets, and subsidiaries are fairly evenly distributed. Table IA.II provides the breakdown of industry composition by acquirer and target SIC codes. Acquirers are concentrated in Pharmaceutical Preparations, with 162 deals having an acquirer belonging to this industry classification. Acquirers and targets share the same four-digit SIC code in slightly over half (107) of deals.

D. Defining Product Markets

Our identification strategy, which we describe in Section III below, hinges on within-deal variation in product market consolidation. We must therefore identify each drug's product market. To do so, we draw on two well-established drug classification systems. Our primary product market definition is based on ATC codes from the WHO. ATC codes are a widely accepted classification system with five nested levels: main anatomical groups (1^{st} level or one-digit), therapeutic subgroups (2^{nd} level or three-digit), pharmacological subgroups (3^{rd} level or four-digit), chemical subgroups (4^{th} level or five-digit), and chemical substance (5^{th} level or seven-digit). Internet Appendix Table IA.III provides an example. We are able to merge ATC codes with NADAC prices using NDCs for 71% of our sample. Our merge rate increases to 87% by filling missing ATC codes with ATC codes of drugs with identical active ingredients from the NDC description. We also infer product markets from pharmacological class categories reported in the NDC directory. These categories (i.e., MoA and EPC) correspond to the listed drug's active moieties. Our baseline overlap measure is acquirer/target three-digit ATC overlap, but we verify robustness to more narrowly defined ATC overlap and MoA/EPC overlap as well.

To address the concern that acquirers stop competition before it begins, we follow Cunningham, Ederer, and Ma (2021) and supplement three-digit ATC overlap with "pipeline" overlap. If a target firm has no drugs listed in the FDA Orange Book database, we identify the type of drugs the target is developing by conducting Google searches for news articles released around the merger announcement date containing the names of the acquirer and target firms. For example, Merck's drug Temodar corresponds to ATC3 code L01 for "Antineoplastic Agents," drugs intended to prevent or slow the progress of a neoplasm (tumor). When Merck acquired Israel-based biopharmaceutical firm cCAM Biotherapeutics in 2015, news reports revealed that the cCAM deal included several early-stage drugs that were similar. In one report, Merck Research Laboratories president Dr. Roger Perlmutter said, "The acquisition of cCAM supports our objective to advance the care of patients with cancer by stimulating tumor-directed immune responses."[14] Because the drug Temodar's ATC code description closely resembles the description of cCAM's pipeline, we assume that Temodar shares a product market with the target firm. Our results, however, are robust to ignoring such pipeline overlap.[15]

[14] "Merck to acquire Israel's cCAM Biotherapeutics for $605m," *Pharmaceutical Technology*, July 29, 2015.

[15] See Internet Appendix Table IA.VII.

III. Drug Price Changes around Pharmaceutical Mergers

This section investigates changes in drug prices around mergers. We begin by describing our empirical strategy, including how we identify drugs whose product markets consolidate around mergers and how we match them with control products. We then motivate our DID model and verify its assumptions. Next, we perform our baseline tests. We conclude by validating the robustness of our results and examining their cross-sectional variation.

A. DID Model and Assumptions

Our empirical strategy for testing how product prices move around mergers is to compare changes in the prices of drugs influenced by mergers to simultaneous price changes of drugs unaffected by a merger that are otherwise similar. Because mergers are not based on random assignment, we must carefully control for the firm and product characteristics that may drive merger activity or affect product pricing. We hold firm characteristics constant by exploiting within-deal variation in product market consolidation. In other words, our sample and control drug products both belong to the same firm (the acquirer).[16] We control for product characteristics via a careful matching process. We elaborate on our identification strategy below.

Figure 5 illustrates how we select sample drugs whose product markets consolidate as a result of the merger. The large circle depicts the drug portfolio of the acquiring firm while the small circle illustrates target-firm drugs. The shaded intersection of these circles represents an overlapping product market. An acquiring firm's drug belongs to our sample if one of the target firm's drugs shares its product market. In our example, acquirer drug A1 is sampled because it shares a product market with target drug T1. We use ATC codes to identify product markets and assume that consolidation occurs in overlapping product markets; an acquiring firm's drug is sampled if its three-digit ATC code corresponds to the same code as at least one of the target firm's drugs.[17] Often target firms do not yet sell drugs. In these mergers, we sample an acquirer drug if its ATC code description closely resembles the description of the target firm's pipeline, as described in Section III.D.

At the time of the merger, the average acquirer manufactures 324 individual products with prices in the NADAC database while the average target manufactures only 26 such drugs. This suggests that, as is typical in mergers, acquirers in our sample tend to be substantially larger than targets. Of the 202 pharmaceutical mergers that occur during our sample period, 85 involve at least one overlapping product market. Because we use within-deal variation in product market consolidation for identification, we must condition on

[16] Section III.D provides justification for our focus on acquirer drugs and discusses robustness to examining target drugs instead.

[17] The majority of drugs in our sample (65%) correspond to only one ATC3 code. If a drug corresponds to more than one code, we sample the drug if at least one of its codes corresponds to at least one of the target drug's codes. Internet Appendix Table IA.IV provides a hypothetical example.

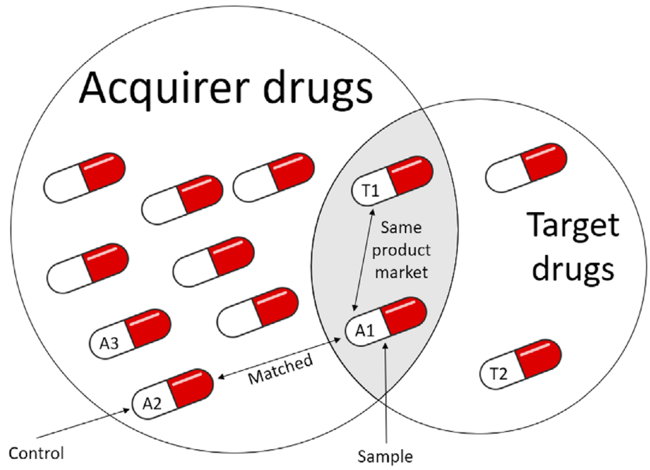

Figure 5. Identifying acquirer/target product market overlap. We examine how product prices change around mergers by exploiting within-deal variation in product market consolidation to compare changes in prices of sampled drugs whose product markets consolidate due to the merger to simultaneous price changes of similar control products. This figure illustrates how we select sample and control drugs. The large circle depicts the drug portfolio of acquiring firm A while the small circle illustrates drugs belonging to target firm T. Their shaded intersection represents an overlapping product market, shared by acquirer drug A1 and target drug T1. Potential control drugs are acquirer drugs whose product markets do not overlap with any of the target's product markets (A2, A3, etc.). We match sample drug A1 to these potential controls on brand name versus generic, prescription versus over-the-counter, and patent and exclusivity rights statuses, then select the drug with the closest total units reimbursed by Medicaid last quarter, represented by drug A2 in this example. (Color figure can be viewed at wileyonlinelibrary.com)

deals with overlapping product markets. Within the remaining deals, 23.6% of acquirer drugs overlap with target product markets; 90.6% of the overlap is attributable to target drugs already in distribution, while the remaining 9.4% of overlapping target drugs are still in the development pipeline.

We next must select appropriate control drugs. We limit potential controls to the pool of acquirer drugs without product market overlap. This allows us to exploit variation in market power consolidation while holding firm characteristics constant. Controlling for drug-level characteristics is also important. We match each drug in our sample to an acquirer drug unaffected by the merger along the following dimensions: brand name versus generic, prescription versus over-the-counter, and patent and exclusivity rights status. We then select the drug with the closest total units reimbursed by Medicaid last quarter, a proxy for lagged demand. If we obtain more than one match, we select the drug closest in unit price. In Figure 5, sample drug A1 is matched with control drug A2, and we compare changes in the price of drug A1 around the merger to simultaneous changes in the price of drug A2.

Table II reproduces drug-level summary statistics for acquirer drugs the quarter prior to the acquisition. We then summarize our DID sample.

Table II

Difference-in-Differences (DID) Sample Summary Statistics

This table presents drug characteristics for our DID sample. We first summarize all acquirer drugs the quarter prior to the acquisition. We then summarize the sample of acquirer drugs whose product market consolidates as a result of the merger. We match each sample drug with a control drug produced by the same acquiring firm but without target overlap. We select control drugs with identical characteristics (brand name versus generic, prescription versus over-the-counter, under patent, under exclusivity) and with the closest total units reimbursed the quarter of the acquisition. *Brand name*, *Generic*, *Prescription*, and *Over-the-counter* are indicator variables for drug characteristics created from the NADAC database. *Patent* (*Exclusivity*) is an indicator equal to one if the drug is under patent (exclusivity protected), as noted in the Food and Drug Administration (FDA) database. *Total units reimbursed (millions)* are the total units reimbursed in the prior quarter as reported in Medicaid's State Drug Utilization Data (SDUD).

		DID Sample			
	All Acquirer Drugs	Sample	Controls	Sample - Control	
	(1)	(2)	(3)	Difference	*t*-stat
Price	14.191	10.874	8.072	2.802	2.489
Brand name	0.270	0.215	0.215	0	N/A
Generic	0.730	0.785	0.785	0	N/A
Prescription	0.847	0.895	0.895	0	N/A
Over-the-counter	0.153	0.105	0.105	0	N/A
Patent	0.135	0.098	0.098	0	N/A
Exclusivity	0.038	0.025	0.025	0	N/A
Total units reimbursed (millions)	0.334	0.426	0.359	0.067	1.846

Relative to the full NADAC drug sample presented in Table I, acquirer drugs tend to be more expensive. Acquirer drugs are also more likely to be brand name and covered under patents or exclusivity rights. More importantly for our empirical strategy, sampled acquirer drugs with product market overlap are different: They are cheaper than the average acquirer drug and more likely to be generic and prescription but less likely to be patented or associated with exclusivity rights.[18] Although descriptive in nature, this table points to the importance of carefully selecting control drugs. Because we exactly match sample and control drugs on brand name versus generic, prescription versus over-the-counter, patent and exclusivity rights status, differences in the sample and control groups along these dimensions equal zero, as shown. Sample and control drugs are also statistically and economically similar in terms of reimburse-

[18] Multivariate regressions in Internet Appendix Table IA.V generally confirm these univariate findings: Relative to other acquirer drugs, acquirer drugs whose product market overlaps with target-firm drugs are more likely to be generic and prescription. Drug-level and product market–level competition metrics are mixed: Regressions show that sample drugs are more likely to be patented but are less likely to be covered under exclusivity rights and that they come from product markets (ATC3 codes) that are less concentrated (with more labelers) but do not face generic competition. These drugs also tend to have decreased in price slightly relative to other acquirer drugs in recent quarters. Finally, multiuse drugs associated with multiple ATC3 codes are not sampled significantly more often.

ment rates. Thus, the only meaningful difference between sample and control drugs is price level (difference = \$2.80, t-statistic = 2.49). We do not explicitly match on price levels because imposing price bands around controls restricts our pool of potential controls even further and could bias DID coefficient estimates.[19] This slight baseline difference nonetheless emphasizes the importance of controlling for price levels by examining within-product changes in (log) price, which we do.

After successfully matching sample and control drugs along observables, we use the following DID equation to remove common time trends:

$$ln(Price_{i,k,t}) = \theta Consolidate_{i,k} * Post_{k,t} + \delta_{i,k} + \eta_{k,t} + \epsilon_{i,k,t}, \tag{1}$$

where $Price_{i,k,t}$ is the price of drug i produced by the acquiring firm in deal k at event quarter t, *Consolidate* equals one if the drug belongs to a product market that is consolidating due to the merger and zero for matched control drugs, and *Post* equals zero before deal k and one during and after the merger announcement quarter. Our coefficient of interest is θ, which is associated with the interaction term between *Consolidate* and *Post*. This coefficient captures the relative difference in log price from the premerger to postmerger periods (first difference) between drugs directly affected by the merger and control products (second difference). Put differently, it is the difference in the approximate percentage price change of drugs subject to product market consolidation resulting from the merger and similar drugs produced by the same manufacturer conducting the acquisition that are not subject to product market consolidation. As our DID sample is constructed at the deal level, we follow the recommendations of prior studies (e.g., Gormley and Matsa (2011), Sun and Abraham (2021), Baker, Larcker, and Wang (2022)) to "saturate" product and time fixed effects with deal indicators: $\delta_{i,k}$ denotes drug-deal fixed effects, which control for time-invariant differences in price levels across products, and $\eta_{k,t}$ denotes deal-event-time fixed effects, which net out time-invariant firm (drug manufacturer) characteristics as well as simultaneous price changes of matched control drugs. We present t-statistics based on robust standard errors clustered at the product level.

A key identifying assumption when using the DID approach is parallel trends. This assumption requires that, in the absence of a merger, the difference in prices between sample and control drugs would have been constant over time. To provide support for this assumption, we estimate:

$$Ln(Price_{i,k,t}) = \sum_{\substack{t=-6 \\ t\neq -1}}^{+6} \theta_t Consolidate_{i,k} * Qtr_{k,t} + \delta_{i,k} + \eta_{k,t} + \epsilon_{i,k,t}. \tag{2}$$

This equation is equivalent to our baseline DID model, except we replace the postmerger indicator with multiple indicators for event time: $Qtr_{k,t}$ equals one

[19] In our setting, for instance, we may inadvertently select controls with recent price increases that experience (downward) reversion to the mean in the near future, which would bias our DID coefficient estimate upward.

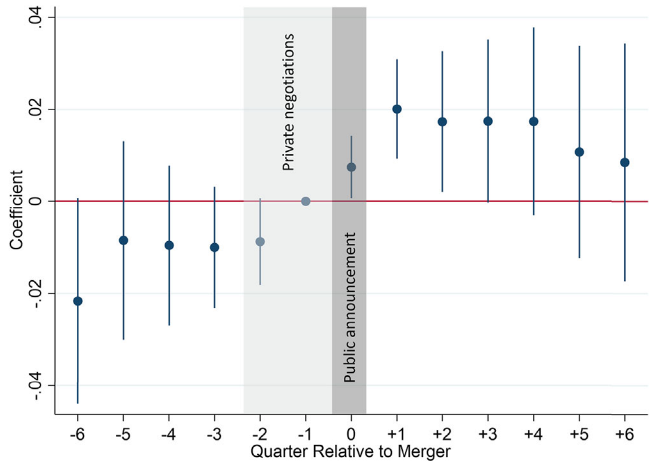

Figure 6. Drug price trends around merger announcements. This figure plots price dynamics of sample drugs whose product markets consolidate around the merger event, relative to control drugs. The plot depicts coefficient estimates of θ_t and their 95% confidence intervals from the following difference-in-differences regression:

$$Ln(Price_{i,k,t}) = \sum_{\substack{t=-6 \\ t\neq -1}}^{+6} \theta_t Consolidate_{i,k} * Qtr_{k,t} + \delta_{i,k} + \eta_{k,t} + \epsilon_{i,k,t}.$$

The dependent variable is the natural logarithm of the inflation-adjusted price of pharmaceutical product i, associated with deal k, averaged over quarter t. *Consolidate* is an indicator equal to one for acquirer drugs whose product market overlaps with a target firm product market. We match each sample drug with a similar control drug produced by the same firm, as detailed in Section III. $Qtr_{k,t}$ is an indicator that equals one if the observation corresponds to event quarter t for deal k. $\delta_{i,k}$ and $\eta_{k,t}$ represent product (NDC)-deal and deal-event-time fixed effects. The dark shaded region illustrates the public merger announcement period (quarter 0) and the light shaded region illustrates when private negotiations typically take place (quarters -1 and -2).
(Color figure can be viewed at wileyonlinelibrary.com)

if the observation corresponds to quarter t relative to the announcement quarter for deal k. We use one quarter prior to the merger as the base quarter.

For the parallel trends assumption, we are interested in whether the coefficients associated with the interactions between consolidating product markets and premerger event times are significantly different from zero. Figure 6 plots estimates of differences in price trends between sample and control drugs around mergers. All premerger θ coefficients are statistically indistinguishable from zero, consistent with parallel trends.

One potential concern is that drugs in consolidating product markets appear to increase in price relative to control drugs as early as the quarter prior to the merger. The mostly likely explanation for these anticipatory effects is that corporate insiders charged with product pricing decisions are aware of the merger well before its public announcement. In fact, a four-month lag is typical between initial rumors and merger announcements;[20] industry experts report a six-month lag from merger inception to consummation is not uncommon.[21] Rambachan and Roth (2023) claim that anticipatory effects are a common issue in parallel trends analysis and propose renormalizing the definition of pretreatment period. Noting that our time cutoff likely produces conservative estimates, we retain the merger quarter as our baseline time threshold. Nevertheless, we confirm robustness to renormalizing our pre/post cutoff to including the merger negotiation period in Figure IA.1 and Table IA.VIII and to defining the event time relative to merger rumors in Figure IA.2 and Table IA.IX.

In addition to confirming the validity of the parallel trends assumption, Figure 6 informs us about the changes in prices around mergers. Prices of drugs experiencing product market consolidation increase about 2% more than control drugs at the time of the merger and remain significantly higher for four quarters. Taken as a whole, these findings indicate that there are few significant differences in price movements between sample and control drugs in the preperiod (consistent with parallel trends) but provide evidence that drugs affected by mergers increase in price significantly more at the time of the merger and remain elevated for almost one year.

B. DID Regressions Results and Discussion

Table III presents our DID analyses with varying windows around the merger quarter. Regardless of the number of quarters around the merger we include, we observe positive and significant coefficients on the consolidation/postmerger interaction term. Our estimates range from 1.3% using the period spanning from one quarter prior to the merger until one quarter after the merger to 2.2% when we focus on four to six quarters pre- and postmerger. These findings suggest that prices of drugs whose product markets are directly impacted by the merger increase significantly more than those of matched control drugs.[22] Although synergy gains may also be realized within consolidating product markets, overall the prevailing price change within product markets consolidating around mergers is positive. This evidence supports the *Market Power Hypothesis* over the *Synergistic Gains Hypothesis*.

According to the Federal Trade Commission, when approving a merger "the key question the Agency asks is whether the proposed merger is likely to

[20] The median number of days between SDC's Original Date Announced and Date Announced for the subsample of pharmaceutical mergers flagged as rumored deals and with nonzero differences between inferred rumor date and announcement date is 127 days.

[21] "What You Need To Know About Mergers & Acquisitions: 12 Key Considerations When Selling Your Company." *Forbes*. August 27, 2018.

[22] These differences are not driven by a decline in control drug prices. Rather, drug prices increase for both sample and control drugs but more so for sample drugs. Using a balanced sample of drugs with prices available from six quarters before the merger to six quarters after, we find that average postmerger prices are 5.8% greater than premerger prices for sample drugs and 4.0% greater for control drugs. As a point of reference, the analogous inflation-adjusted price increase for drugs that are neither sample nor control drugs is 2.7%.

Table III

Mergers and Drug Prices

This table presents difference-in-differences regressions of acquirer drug prices on merger activity. We exploit within-deal variation in product market consolidation around mergers to compare prices of sample drugs and matched control drugs. The dependent variable is *Ln(Price)*, which equals the natural log of the National Average Drug Acquisition Cost (NADAC) per unit, averaged over the quarter and expressed in 2019 dollars. *Consolidate* is an indicator that equals one for acquirer drugs with target firm overlap and zero for matched control drugs. Overlap implies that the target firm produces a drug sharing the same ATC3 code or has similar drugs in the production pipeline. Each sample drug is matched with a control drug produced by the same manufacturer, with identical characteristics (brand name versus generic, prescription versus over-the-counter, under patent, under exclusivity), closest in total units reimbursed the quarter of the acquisition. If multiple control drugs have the same total units reimbursed, we match with the drug closest in unit price. *Post* is an indicator that equals one for the quarters of and after the merger. We vary the number of quarters around the merger, as noted. We include product (NDC)-deal and deal-event-time fixed effects. *t*-statistics based on robust standard errors clustered at the product level are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	±1 qtr (1)	±2 qtr (2)	±3 qtr (3)	±4 qtr (4)	±5 qtr (5)	±6 qtr (6)
Consolidate × Post	0.013***	0.018***	0.020***	0.022***	0.022**	0.022**
	(3.144)	(3.239)	(3.070)	(2.863)	(2.499)	(2.275)
Fixed effects	Product (NDC)-Deal, Deal-Event Time					
Observations	41,609	68,514	93,857	118,159	141,683	163,787
Adjusted R^2	0.999	0.997	0.996	0.995	0.994	0.993

create or enhance market power or facilitate its exercise."[23] How, then, do we find price increases associated with deals approved by regulators? Several factors could come into play. The first is the magnitude of the price increase. While the effects we document are statistically significant, their economic magnitude may be insufficient to garner the attention of regulators. An additional point of consideration in the merger approval process is the time it would take a competitor to enter the market, or "timeliness of entry." Entry is generally considered "timely" if achieved within two years, and the Agency challenges mergers only if they determine that entry by competitors would not be timely. Our time trends results in Figure 6 show that price increases are sustained only for three to four quarters postmerger. A final reason these mergers may be approved relates to the difficulty of predicting demand within the pharmaceutical industry. Because pharmaceutical products have high switching costs (e.g., due to side effects), they are likely associated with low price elasticity and low demand-side substitutability. Prices may therefore remain inflated longer than they would in other industries, making it particularly difficult to predict price changes around mergers within this industry. Put differently, as researchers, we have the benefit of hindsight.

[23] See https://www.ftc.gov/tips-advice/competition-guidance/guide-antitrust-laws/mergers.

C. DID Robustness Checks

This section verifies the robustness of our baseline DID analyses to several alternative specifications. We begin by altering our definition of acquirer/target overlap, which is how we identify drugs with product market consolidation. We originally define overlap as cases in which either the target firm produces a drug with the same three-digit ATC code or the target firm's pipeline includes products with similar descriptions as the ATC code descriptions. ATC codes are seven-digit codes representing five nested levels of classifications: main anatomical groups (1^{st} or one-digit level), therapeutic subgroups (2^{nd} or three-digit level), pharmacological subgroups (3^{rd} or four-digit level), chemical subgroups (4^{th} or five-digit level), and the chemical substance (5^{th} or seven-digit level).[24]

Models (1) to (6) in Table IV condition on same four-digit, five-digit, or seven-digit ATC code acquirer/target overlap. For brevity, we restrict our time windows to correspond to plus or minus four or six quarters around the merger. Because conditioning on higher degrees of ATC code matching is increasingly restrictive, the number of drugs we identify with product market consolidation falls, with the sample size decreasing accordingly. Nonetheless, in all cases, our coefficients of interest are positive and significant. Indeed, they are slightly larger than coefficients in our baseline regressions and increase in product similarity. Coefficient estimates range from 0.025 to 0.037 four quarters pre- and postmerger and from 0.027 to 0.041 six quarters pre- and postmerger. These estimates imply that acquirer drugs with target product market overlap increase in price significantly more than control drugs, even—and especially—when product markets are more narrowly defined.

Models (7) and (8) of Table IV use an alternative definition of overlap. Hammoudeh and Nain (2020) and Cunningham, Ederer, and Ma (2021) infer acquirer/target overlap using MoA or EPC.[25] To verify robustness to this alternative product market definition, we consider acquirer and target product markets as overlapping if they share an MoA or EPC. While MoA/EPC overlap almost always implies ATC3 overlap (96% of drugs with MoA/EPC overlap also share an ATC3 code with a target drug), only approximately half of ATC3 overlaps also have MoA/EPC overlap. This implies that drugs with MoA/EPC overlap roughly represent a subset of drugs with ATC3 code overlap. Within this subset, our results not only hold but are stronger. Acquirer drugs with an MoA or EPC equivalent to the MoA or EPC of a target drug experience 3.0% to 3.1% larger increases in price than matched control drugs.

We continue our robustness checks in Table V by examining price changes around mergers within "cleaner" samples and control pools. We begin by examining how our choice of control drugs impacts our findings. Our baseline results allow drugs with product market overlap in one merger to serve as control

[24] We present an example in Internet Appendix Table IA.III.

[25] Hammoudeh and Nain (2020) use textual analysis to calculate similarity scores between descriptions of therapeutic area and MoA, whereas Cunningham, Ederer, and Ma (2021) directly match drugs on therapeutic areas and MoA, as we do.

Table IV

Alternative Classifications of Acquirer-Target Overlap

This table examines the robustness of the Table III difference-in-differences regressions of drug prices on merger activity to alternative classifications of acquirer-target product market overlap. Our original overlap classification conditions on at least one 2nd (three-digit) level ATC code of the acquirer drug matching at least one 2nd (three-digit) level ATC code of a target drug. Three-digit ATC codes correspond to therapeutic subgroup. We first show robustness to the 3rd (four-digit), 4th (five-digit), and 5th (seven-digit) classification levels, which correspond to the pharmacological subgroup, chemical subgroup, and chemical substance, respectively. We provide examples of ATC code hierarchy and overlap definitions in Internet Appendix Table IA.III. We then show robustness to matching on mechanism of action (MoA) or established pharmacologic class (EPC). t-statistics based on robust standard errors clustered at the product level are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	4-digit ATC code		5-digit ATC code		7-digit ATC code		MoA or EPC	
	±4 qtr (1)	±6 qtr (2)	±4 qtr (3)	±6 qtr (4)	±4 qtr (5)	±6 qtr (6)	±4 qtr (7)	±6 qtr (8)
Consolidate* × Post	0.025***	0.027***	0.035***	0.040***	0.037***	0.041***	0.030***	0.031**
	(3.632)	(2.947)	(4.915)	(4.229)	(4.125)	(3.539)	(2.998)	(2.307)
Fixed effects	Product (NDC)-Deal, Deal-Event Time							
Observations	94,160	130,885	71,373	99,443	45,125	62,870	53,216	74,504
Adjusted R^2	0.995	0.993	0.995	0.993	0.996	0.994	0.992	0.988

Table V

Robustness of Mergers and Drug Prices to Alternative Samples and Controls

This table examines the robustness of the Table III difference-in-differences regressions of drug prices on merger activity. Models (1) and (2) limit the pool of control drugs to acquirer drugs with no product market overlap (i.e., not sampled for another merger) within six quarters of the merger event. Models (3) and (4) limit the pool of control drugs to acquirer drugs with no product market overlap during our entire sample period. Models (5) and (6) limit the pool of control drugs to acquirer drugs with no product market overlap within six quarters of the merger event and sample drugs to drugs with no product market overlap from another merger within six quarters of the merger event. Models (7) and (8) limit the pool of control drugs to acquirer drugs with no product market overlap during our entire sample period and sample drugs to drugs with no product market overlap from another merger during the entire sample period. t-statistics based on robust standard errors clustered at the product level are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	Baseline		Baseline		No Other Overlap within 6 qtr		No Other Overlap	
	No Overlap within 6 qtr		No Overlap		No overlap within 6 qtr		No Overlap	
Sample: Control Group:	±4 qtr (1)	±6 qtr (2)	±4 qtr (3)	±6 qtr (4)	±4 qtr (5)	±6 qtr (6)	±4 qtr (7)	±6 qtr (8)
Consolidate × Post	0.016	0.016	0.016	0.016	0.036***	0.034***	0.035***	0.034***
	(1.548)	(1.346)	(1.570)	(1.369)	(3.921)	(2.880)	(3.711)	(2.743)
Fixed effects	Product (NDC)-Deal, Deal-Event Time							
Observations	116,107	160,937	116,119	160,948	75,235	106,027	75,130	105,837
Adjusted R^2	0.995	0.993	0.995	0.993	0.993	0.990	0.993	0.990

drugs in other mergers. The main benefit to this approach is that drugs with overlap likely tend to be more similar to one another than to drugs that never experience product market consolidation. In general, broadening the pool of potential controls allows for better matching. Nonetheless, this approach may be problematic if mergers occur in quick succession. To see how our control sample impacts our findings, in Models (1) and (2) we limit potential control drugs to acquirer drugs that do not experience product market overlap with a target firm of another merger within six quarters. Models (3) and (4) then restrict the pool of controls further to the subsample of drugs that never experience overlap during our entire sample period. That is, these control drugs are never sampled themselves. Restricting controls results in a slight drop in sample size as we are now unable to identify controls for several drugs. Models (1) to (4) continue to show positive coefficients of interest, although statistically insignificant and somewhat lower in magnitude compared to those found in our baseline models. Estimates of incremental price increases for four or six quarters around the merger are consistently 1.6%.

We next restrict *sample* drugs (in addition to control drugs) to these cleaner subsamples. Models (5) and (6) restrict both sample and control drugs to drugs not sampled in other mergers within six quarters. Models (7) and (8) limit both sample and control drugs to drugs not sampled in any other merger, that is, we match our subsample of acquirer drugs that experienced product market consolidation only once with controls never directly affected by a merger. In models (5) to (8), our sample size drops as expected, but our coefficients of interest rise. The coefficients from the models using four quarters pre- and postmerger increase from our baseline estimate of 0.022 to 0.035 and 0.036. The coefficients in models with a longer time window increase similarly. We infer from these robustness checks that our choices of sample and control drugs do not drive our main finding of prices rising around mergers. These robustness checks produce estimates of price effects ranging from 1.6% to 3.6%, which encompasses our original estimate of 2.2%.

D. Drug Price Changes within the Target Firm

The results above show a robust and consistent finding: Acquirer drugs that share a product market with the target firm tend to increase in price around mergers more than similar acquirer drugs with no product market overlap. A natural question that arises is whether prices of drugs originally produced by the *target* firm follow a similar pattern. This subsection provides further justification for our focus on acquirer drugs and then discusses the extent to which our results extend to target drugs.

Our primary analysis focuses on acquirer drugs for two reasons. First, acquirer drugs yield a larger sample than target drugs. As mentioned above, at the time of the merger the average acquiring firm sells over 12 times as many drugs as the average target. Further, many target drugs are still in development at the time of the merger and hence have no price data. This smaller sample of drugs leads to less powerful tests and makes matching target drugs with product market overlap to potential control target drugs challenging. In fact, over one-quarter of targets produce two or fewer drugs, implying zero or at most one potential matched control.

The second reason we hesitate to study target drugs is that the parallel trends assumption necessary for our DID model is violated. Relative to matched controls, prices of target drugs with acquirer overlap steadily increase the year prior to the merger.

Keeping in mind the important caveats regarding sample size and nonparallel pre-event price trends, we run DID regressions analogous to those in Table III, substituting target drugs for acquirer drugs. We tabulate these results in Internet Appendix Table IA.X. Around mergers, target drugs whose product market overlaps with the acquirer increase in price significantly more than matched control drugs. These findings are consistent with our main takeaway that merger-induced market power gains facilitate price increases.

We present these results for completeness but note that they should be interpreted with caution. Because the parallel trends assumption is violated, we cannot confidently assume that sample and control drug prices would have trended in the same direction in the absence of the merger. Price changes of matched control drugs are unlikely to represent a valid counterfactual. In addition, while the economic magnitude of the price effect appears to grow over time (from 2.8% one quarter around the merger to 14.5% six quarters pre- and postmerger), this growth is due to premerger prices being increasingly lower as we expand the premerger period, not to postmerger prices rising over time. In sum, while all evidence points to affected target drugs' prices also increasing around mergers—with the most conservative estimates even implying a similar magnitude (2.8%) as the change in acquirer drug prices—these findings should be interpreted with caution.

E. Competition and Drug Prices around Mergers

This section uses cross-sectional tests to strengthen identification and explore the mechanisms behind the link between mergers and price increases. Specifically, we test whether price increases are larger within concentrated (less competitive) product markets. We expect price increases to be larger within concentrated product markets because switching costs are likely higher as there are fewer substitutes. Further, because concentrated markets have fewer players, in expectation, the combination of two players has a greater impact on the distribution of market share within a concentrated market than within a disperse market with many players. In addition to bolstering our argument that the link between mergers and drug prices is not a spurious correlation, this cross-sectional analysis provides important insights for regulators scrutinizing these deals.

To test our prediction, we augment our baseline DID regressions of log prices with the indicator *Concentrated*, which equals one if the drug belongs to a concentrated product market. We assume that a market is concentrated if the number of unique labelers in the product market the year before the merger falls below the median. If a drug belongs to more than one product market (ATC3 code), we associate it with the least competitive market. When we interact *Concentrated* with our indicators for product market consolidation and the postmerger period, we can interpret the coefficient on the triple interaction as the difference in price change around mergers between sample and control drugs in concentrated versus disperse product markets.

Models (1) and (2) of Table VI show that prices increase more within concentrated product markets than disperse product markets. Concentrated product markets that experience further consolidation due to a merger are associated with price increases around 5% greater than consolidating product markets with more competitors. These findings are consistent with our prediction that merger-induced market power consolidation allows prices to rise and provide further support for the *Market Power Hypothesis*.

Our second cross-sectional test is motivated by the observation that pricing power should be easier to establish in the absence of generic competition. We therefore create the indicator *No generic*, which equals one if the drug faces no generic competition in the same ATC7 code space. About one-quarter (26%) of drugs in our DID sample have no generic competition. Models (3) and (4) of Table VI document coefficients on the triple interaction of *No generic*, *Consolidate*, and *Post* equal to 0.057 and 0.104, respectively. As expected, drugs in less competitive product markets—those without generic competition—increase in price more around mergers than other drugs experiencing product market consolidation.

Cunningham, Ederer, and Ma (2021) examine so-called "killer acquisitions" in which firms acquire competitors to halt development. Using data from the pharmaceutical industry, they show that acquirers are more likely to halt the development of target firm drugs if the acquirer shares the product market. Further, drugs are less likely to advance from Phase 1 to Phase 2 if they are acquired by a firm competing in the same product market. The authors also show that "killer" deals disproportionately occur beneath regulatory thresholds, thereby skirting scrutiny. If deals that occur earlier—when the target's drug is in the development as opposed to production phase—are more likely to suppress competition, then price changes around these deals may be greater. We test this prediction by adding an indicator for pipeline overlap to our DID regressions; *Pipeline* equals one if the drug shares a product market with drugs in the target firm's pipeline. Because this indicator can only equal one for drugs with overlapping product markets, the coefficient associated with the triple interaction term *Pipeline* $\times$ *Consolidate* $\times$ *Post* can be interpreted as the incremental price change associated with pipeline overlap. We therefore do not need (and cannot specify a model with) an independent *Pipeline* indicator.

Models (5) and (6) of Table VI report that pipeline overlap is not associated with greater price increases. Although acquisitions curb the production of future products (as Cunningham, Ederer, and Ma (2021) show), we find that acquisitions potentially motivated by suppressing future competition are not associated with larger price increases for existing products, at least not in the short run.

Table VI

Cross-Sectional Variation in Price Changes around Mergers

This table presents triple difference-in-differences regressions of drug prices on merger activity, based on Table III baseline models with drug level interactions. *Concentrated* is an indicator equal to one for concentrated product markets, that is, if the number of labelers in the product market the year before the merger falls below the median. If a drug is associated with multiple ATC3 codes, we assign it to the code with the fewest labelers. *No generic* equals one if the drug faces no generic competition in the same ATC7 code space. *Pipeline* equals one if the overlap between the drug and the target firm's drug portfolio stems from a drug in the pipeline rather than a drug that has already entered the market. t-statistics based on robust standard errors clustered at the product level are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	Concentrated Product Market		No Generic Competition		Overlap from Drug Pipeline	
	±4 qtr (1)	±6 qtr (2)	±4 qtr (3)	±6 qtr (4)	±4 qtr (5)	±6 qtr (6)
Concentrated × Consolidate × Post	0.049***	0.051***				
	(3.418)	(2.780)				
No generic × Consolidate × Post			0.057***	0.104***		
			(3.272)	(4.472)		
Pipeline × Consolidate × Post					−0.014	−0.011
					(−0.974)	(−0.642)
Concentrated × Post	−0.025*	−0.026				
	(−1.951)	(−1.571)				
No generic × Post			0.016	0.012		
			(1.147)	(0.701)		
Consolidate × Post	0.005	0.004	0.026***	0.023**	0.024***	0.023**
	(0.447)	(0.301)	(3.277)	(2.168)	(2.804)	(2.189)
Fixed effects	Product (NDC)-Deal, Deal-Event Time					
Observations	118,159	163,787	118,159	163,787	118,159	163,787
Adjusted R^2	0.995	0.993	0.995	0.993	0.995	0.993

Table VII

Acquirer Returns around Merger Announcements

This table presents five-day cumulative abnormal returns (CARs) to acquirers around M&A announcements. Panel A examines the full sample of pharmaceutical mergers with available drug price and stock price data. *Horizontal* (*Diversifying*) deals involve an acquirer and a target with the same (different) four-digit SIC code. *High* (*Low*) *% overlap* denotes deals with above-average (below-average) *% overlap*, defined as the percentage of acquirer drug products with target firm overlap. Panel B conditions on the subsample of deals with nonzero product market overlap between the acquirer and target and segments the sample on *$ overlap*, the total reimbursement amount of acquirer overlapping drugs over the prior year, scaled by acquirer size (total assets). t-statistics are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Full sample of pharmaceutical mergers

All Deals	Horizontal	Diversifying	Horizontal - Diversifying	High % overlap	Low % overlap	High - Low
0.015***	0.029***	0.002	0.027***	0.034***	0.011**	0.023*
(3.11)	(3.41)	(0.38)	(2.79)	(3.21)	(1.95)	(1.85)

Panel B: Subsample of pharmaceutical mergers with overlap

Sample split:	Mean	Median	75^{th} percentile	90^{th} percentile
High	0.034*	0.024**	0.034*	0.056**
$ overlap	(1.91)	(2.49)	(2.07)	(2.36)
Low	0.011**	0.009	0.010**	0.011**
$ overlap	(2.38)	(1.66)	(2.22)	(2.21)
High -	0.023*	0.015	0.024*	0.045***
Low	(1.83)	(1.35)	(1.97)	(2.72)

IV. Shareholder Value Creation

In this section, we examine the stock market's reactions to pharmaceutical mergers. Our first goal is to quantify potential benefits associated with these mergers by clarifying whether these transactions generate value for shareholders on average. Our second goal is to examine whether shareholder value creation varies with market concentration. If mergers generate shareholder value through synergistic cost-cutting like eliminating redundancies in administration or streamlining distribution channels, then we would not expect merger announcement returns to vary with proxies for concentration. Alternatively, if mergers tend to benefit shareholders because they increase market power, then we would expect higher stock returns around announcements of mergers that concentrate market power more.

Table VII presents merger announcement CARs. Panel A examines the full sample of pharmaceutical mergers. Following Harford (1999), we calculate acquiring firms' CARs by constructing a market model using the CRSP value-weighted market returns over the 200-day period ending 11 days before the merger announcement. We observe positive five-day CARs of 1.5% that are statistically different from zero at the 1% level. These CARs indicate mergers in our sample create shareholder value on average.

Although all of our deals take place within the pharmaceutical industry, there is heterogeneity in the extent to which acquirers and targets are similar. If mergers tend to benefit shareholders through gains in market power, then deals involving more similar acquirers and targets should generate more shareholder value. Because announcement returns correspond to the entire firm, not just individual products, we construct proxies of acquirer-target similarity at the deal level. Our first proxy distinguishes between horizontal and diversifying deals. We define *horizontal* deals as those whose acquirer and target share the same four-digit SIC code. In contrast, *diversifying* deals join firms with different four-digit SIC codes. Industry composition analysis in Internet Appendix Table IA.II reveals that our deals are fairly evenly distributed between horizontal and diversifying deals. Consistent with shareholders of acquiring firms benefiting more from deals that concentrate market power to a greater extent, five-day CARs around horizonal mergers are positive at 2.9% and significantly greater than CARs around diversifying deals, which are statistically and economically indistinguishable from zero. Our second proxy captures the extent to which the acquirer's drugs overlap with the target's drugs. For each drug, we create an indicator equal to one if the target firm produces a drug with the same three-digit ATC code or if the target firm's pipeline includes similar products. We then average this indicator at the deal level to capture the percentage of drugs with overlap and compute the average percent overlap across all deals. The measure *High (Low) % overlap* denotes mergers with above-average (below-average) overlap. Mergers with high overlap are associated with CARs of 3.4%. In contrast, low-overlap deals are associated with significantly lower abnormal returns of 1.1%. Overall, our evidence is consistent with mergers that consolidate market power most generating significantly more wealth for shareholders.

In Panel B, we condition on deals with nonzero acquirer/target product market overlap and refine our overlap measure to reflect the economic importance of the overlap. This sample mirrors our DID merger sample. To gauge the economic importance of overlap, we sum the total dollars reimbursed by Medicaid in the prior year across all acquirer drugs sharing a three-digit ATC code with the target firm. We then scale this aggregate reimbursement amount of overlapping drugs by firm size. *High (Low) $ overlap* denotes mergers whose economic importance of drug overlap falls above (below) various thresholds, as noted. We find that mergers associated with above-average overlap tend to create more wealth for the acquiring firm's shareholders: Five-day CARs average 3.7% for deals associated with above-average overlap but only 1.1% for other deals. We also split the sample along the median, 75$^{\text{th}}$ percentile, and 90$^{\text{th}}$ percentile. As the threshold for the economic importance of overlap increases, so does the market's reaction to the merger announcement and the difference in returns between high- and low-overlap subsets. In sum, merger announcement returns increase in the economic importance of acquirer-target product market overlap.

V. Innovation

It may be optimal for pharmaceutical companies to raise prices if the rents foster innovation. In this section we examine whether pharmaceutical mergers spur innovation. That merged companies will develop more new drugs together than separately is an oft-cited advantage to pharmaceutical mergers. Bena and Li (2014) conclude that mergers improve innovation, particularly if the merging firms overlap in technology prior to the merger. If pharmaceutical mergers also improve innovation, consumers and regulators may tolerate the higher prices that they tend to be associated with. Examining innovation around pharmaceutical mergers thus helps clarify potential trade-offs and welfare implications.

A. Innovation DID Model and Assumptions

To investigate pharmaceutical innovation activity around mergers, we again employ a DID model. Unlike prices, innovation occurs at the firm, not the product, level. Accordingly, we now include in our sample all pharmaceutical manufacturers conducting an acquisition in the given year. To identify control firms, we assign each firm-year observation a propensity-to-acquire score using the Harford (1999) model. This model (shown in Internet Appendix Table IA.XI) predicts merger bidding using abnormal returns, sales growth, noncash working capital, leverage, market-to-book, price-to-earnings, size, and year fixed effects. Then, at the deal level, we match each acquiring firm to the nonacquiring firm with the closest predicted acquisition likelihood the year of the deal.

Table VIII predicts results on how well our matching process selects control firms. In particular, it displays t-tests on differences in means across all variables in the Harford (1999) model. Our acquiring and control firms are well balanced. Moreover, they do not differ significantly along any observables related to acquisition likelihood, and they also achieve common support: Control firms' average propensity to acquire is not significantly different from acquiring firms.

To net out common time trends, we employ the following DID regression of innovation from firm j around merger k in year t:

$$Innovation_{j,k,t} = \theta Merger_{j,k} * Post_{k,t} + \gamma_{j,k} + \delta_{k,t} + \epsilon_{j,k,t}, \tag{3}$$

where *Innovation* equals the natural log of one plus the number of FDA new drug applications. The first model uses all FDA new drug applications. We then bifurcate drug applications into initial versus secondary applications. Initial (or "original") applications are first-time new drug applications whereas secondary (or "supplemental") applications are changes to an FDA-approved application. Finally, we create three groups based on submission classification codes (illustrated in Table IA.I): new drug approvals, labeling changes, and all other codes, including missing values. Our coefficient of interest is θ, which

Table VIII

Firm-Level Matching and Common Support

This table confirms that our propensity score matching achieves well-balanced control firms and satisfies the common support condition. At the deal level, we match each acquiring firm with the control nonacquiring firm closest in predicted acquisition likelihood based on the Harford (1999) model the year of the deal. This table presents means and differences in propensity score and the covariates from the acquisition likelihood model. Table IA.XI presents the acquisition likelihood model. Table AI defines our variables.

	Acquirer	Control	Difference	*t*-stat
Sales growth	1.363	0.254	1.110	0.964
Noncash working capital	−0.001	−0.019	0.018	1.676
Leverage	0.310	0.323	−0.013	−0.651
Market-to-book	4.438	3.931	0.506	0.221
Price-to-earnings	12.196	5.893	6.303	1.006
Size	9.623	9.627	−0.004	−0.012
Cash deviation	0.000	0.001	−0.001	−0.901
Avg. abnormal returns	0.085	0.119	−0.034	−1.005
Propensity score	0.366	0.363	−0.005	0.147

compares acquiring firms' changes in innovation around mergers to innovation changes within matched nonacquiring firms at the same time. We expect θ to be positive if mergers spur innovation. We include deal-firm and deal-event-time fixed effects in all models.

The DID approach above depends on the assumption that the difference in innovation between acquirers and control firms would be constant over time in the absence of a merger. To examine this parallel trends assumption, we regress innovation metrics on indicator variables capturing time relative to the merger event and their interaction with an indicator for merging firms,

$$Innovation_{j,k,t} = \sum_{\substack{t=-3 \\ t\neq -1}}^{+3} \theta_t Merger_{j,k} * Year_{k,t} + \gamma_{j,k} + \delta_{k,t} + \epsilon_{j,k,t}. \quad (4)$$

This equation is otherwise equivalent to our main DID equation. The base year is one year before the merger, and we are interested in whether the coefficients associated with the interaction between the indicators for events years and mergers are significantly different from zero during the years prior to the merger year.

Figure 7 graphs the innovation regression coefficients associated with all merger-year interactions, along with their 95% confidence intervals. For all innovation metrics, the confidence intervals associated with the interaction coefficients before the merger contain zero. Hence, we observe no significant difference in premerger innovation trends across acquiring firms and matched control firms and can assume that the parallel trends assumption is satisfied.

These trends also reveal how innovation changes within acquiring firms, relative to matched control firms. While new drug applications increase during the merger year and one year after, these increases appear to be driven

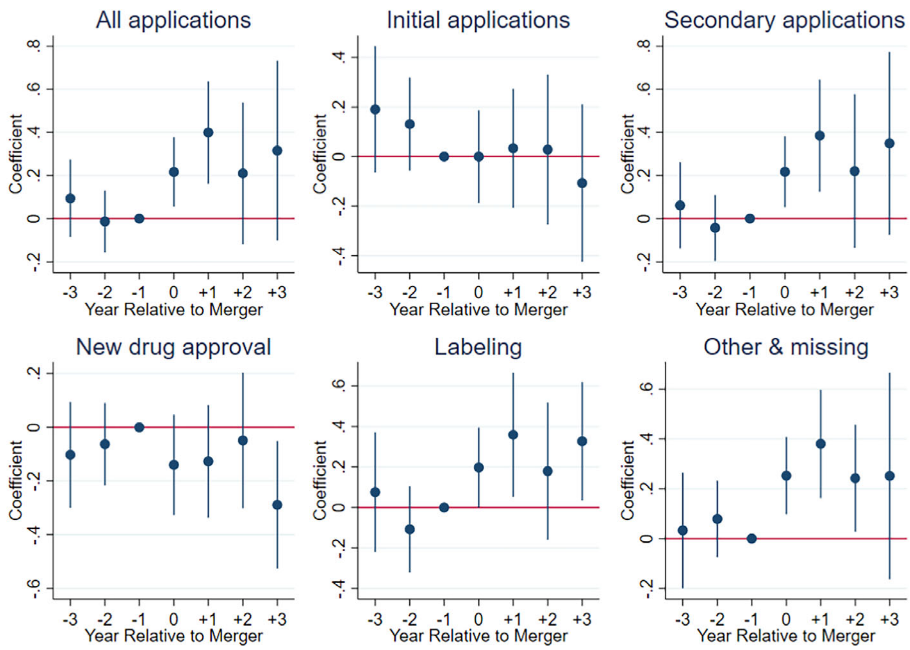

Figure 7. Innovation trends around mergers. This figure tests the parallel trends assumption and plots innovation differences between acquiring and nonacquiring firms. Plots depict coefficients estimates of θ_t and their 95% confidence intervals from difference-in-differences regressions on innovation metrics as follows:

$$Innovation_{j,k,t} = \sum_{\substack{t=-3 \\ t\neq -1}}^{+3} \theta_t Merger_{j,k} * Year_{k,t} + \gamma_{j,k} + \delta_{k,t} + \epsilon_{j,k,t}.$$

The first dependent variable is the natural log of one plus all FDA new drug applications. We then bifurcate drug applications into initial versus secondary applications. Initial (or "original") applications are first-time new drug applications, whereas secondary (or "supplemental") applications are changes to an FDA-approved application. Finally, we create three groups based on submission classification codes (illustrated in Internet Appendix Table IA.I): new drug approvals, labeling changes, and all other codes, including missing values. At the deal level, we match each acquiring firm with the control firm closest in predicted acquisition likelihood from the Harford (1999) model that did not conduct an acquisition the year of the deal. We include the seven years spanning from three years prior to the merger until three years after the merger. $Year_{k,t}$ is an indicator equal to one if the observation corresponds to event year t relative to the merger year. γ_k and $\delta_{k,t}$ represent deal-firm and deal-event-time fixed effects.
(Color figure can be viewed at wileyonlinelibrary.com)

by secondary applications: Secondary applications increase up to 40%, but primary applications remain low. Labeling applications and applications related to other changes such as manufacturing process improvements rise, but not new drug approvals related to new molecules, new active ingredients, or new combinations. We confirm these results in a streamlined regression setting below.

B. Mergers and Innovation

Table IX presents our innovation DID regression analyses. Our models in Panel A examine whether new drug applications increase more for acquiring firms than for matched nonacquiring firms within the seven-year period from three years premerger to three years postmerger. Given that our dependent variable is the natural log of (one plus) the number of applications, we can interpret the coefficient of interest as the approximate incremental percentage change in applications.

Our coefficient of interest in model (1) shows that acquiring firms increase applications 26.7% more than matched nonacquiring firms. Segmenting applications on initial versus secondary submissions in models (2) and (3) and on submission classifications in models (4) to (6), however, shows that the increase in new applications is driven by secondary applications and by applications related to labeling and manufacturing process changes. Initial applications and new drug applications do not increase significantly more for acquiring firms than matched control firms.

Our results in Panel A focus on the average merger in our sample. It may be the case that new drug creation does not meaningfully increase around the average merger but rather only around acquisitions of the most innovative targets. Industry experts argue that biotechnology firms are the most innovative within the pharmaceutical industry, describing innovation as "a force which ostensibly drives the lifeblood of the biopharmaceutical industry."[26] We therefore examine innovation within the subsample of mergers involving a biotech target in Panel B.

To identify biotech targets, we search EDGAR 10-K filings for each acquirer the year of the merger for information about the target firm. In cases in which the target's business operations are unclear from the acquirer's 10-K filing, we supplement with Google searches of the target firm's name. This process yields 15 deals in which the target is a confirmed biotech firm. Panel B of Table IX conditions on this biotech subsample and examines changes in innovation around the merger across acquirers of biotech firms and matched nonacquirers. Our findings for this subsample echo the findings for the full sample. We observe a statistically insignificant though sizable (18.9%) increase in FDA new drug applications, but these excess applications are concentrated within secondary applications related to labeling and other changes. We observe a statistically significant *negative* impact on new drug applications when we form this subsample. These findings suggest that pharmaceutical acquisitions, even of the targets expected to be most innovative ex ante, do not result in meaningful increases in innovation output.

Our innovation analyses above use the same time period as our drug price analysis, which limits us to a short time period beginning in 2013. Yet, new drug development takes time. Accordingly, we next examine the robustness of our innovation results to longer time periods around the merger. We extend

[26] "The most innovative and inventive drug companies of 2022 set the foundation for success before the pandemic," *Fortune*, April 20, 2022.

Table IX
Pharmaceutical Mergers and Innovation

This table presents difference-in-differences regressions of firm-level innovation on merger activity. At the deal level, we match each acquiring firm with the control firm closest in predicted acquisition likelihood that did not conduct an acquisition the year of the deal. Acquisition likelihood is estimated using the Harford (1999) model presented in Internet Appendix Table IA.XI. Panel A includes seven years around the merger. Panel B conditions on the subsample of deals involving a biotech target. Panel C extends our sample to include any Compustat firm that submits an FDA new drug application between 1990 and 2020, and we include the 21 years spanning from 10 years prior to the merger to 10 years after the merger. *Post* corresponds to the years of and after the merger. The dependent variable is the natural log of one plus FDA new drug applications. Model (1) includes all FDA new drug applications. Models (2) and (3) segment on submission type: "Initial" (or "original") applications are first-time new drug applications whereas "secondary" (or "supplemental") applications are changes to an FDA-approved application. Models (4) to (6) segment on submission classification codes, illustrated in Internet Appendix Table IA.I. Model (4) presents new drug approvals, model (5) presents new FDA applications related to labeling changes, and model (6) includes the remaining classification codes (including missing codes). We include deal-firm and deal-event-time fixed effects in all models. t-statistics based on robust standard errors clustered at the firm level are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Three years before and after merger, drug price sample

		Initial vs. Secondary		Submission Classification		
	All Applications (1)	Initial (2)	Secondary (3)	New Drug (4)	Labeling (5)	Other (6)
Merger × Post	0.267**	−0.085	0.287**	−0.095	0.280***	0.254***
	(2.386)	(−0.993)	(2.526)	(−1.656)	(2.774)	(2.770)
Fixed effects	Deal-Firm, Deal-Event Time					
Observations	990	990	990	990	990	990
Adjusted R^2	0.932	0.745	0.921	0.279	0.906	0.875

(*Continued*)

Table IX—*Continued*

Panel B: Three years before and after merger, biotech subsample of drug price sample

		Initial vs. Secondary		Submission Classification		
	All Applications (1)	Initial (2)	Secondary (3)	New Drug (4)	Labeling (5)	Other (6)
Merger × Post	0.189	−0.245	0.210	−0.401**	0.237	0.312
	(1.539)	(−1.109)	(1.648)	(−2.302)	(0.963)	(1.560)
Fixed effects	Deal-Firm, Deal-Event Time					
Observations	136	136	136	136	136	136
Adjusted R^2	0.955	0.774	0.950	0.282	0.896	0.842

Panel C: Ten years before and after merger, extended sample

		Initial vs. Secondary		Submission Classification		
	All Applications (1)	Initial (2)	Secondary (3)	New drug (4)	Labeling (5)	Other (6)
Merger × Post	0.014	−0.039	0.023	−0.027	0.006	0.025
	(0.232)	(−1.355)	(0.393)	(−0.921)	(0.096)	(0.414)
Fixed effects	Deal-Firm, Deal-Event Time					
Observations	15,044	15,044	15,044	15,044	15,044	15,044
Adjusted R^2	0.809	0.660	0.813	0.473	0.820	0.737

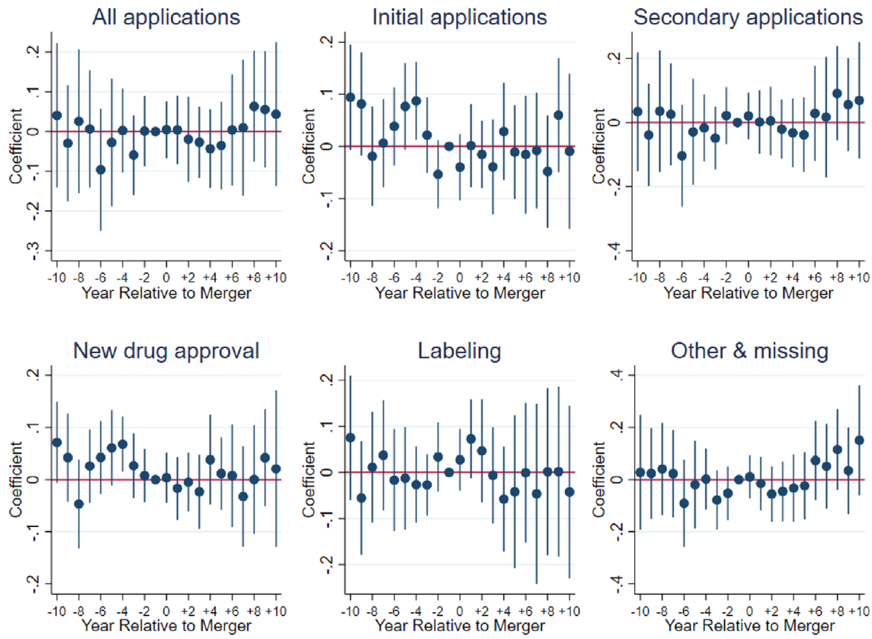

Figure 8. Long-run innovation trends around mergers. This figure plots innovation differences between acquiring and nonacquiring firms. We extend our sample to include any firm that submits an FDA new drug application between 1990 and 2020 with available Compustat data. We include the 21 years spanning from 10 years prior to the merger to 10 years after the merger. Otherwise, our methodology and model are identical to those discussed and presented in Figure 7. (Color figure can be viewed at wileyonlinelibrary.com)

our sample of merging firms and potential control firms to include all FDA new drug applicants between 1990 and 2020. We then merge this larger and longer sample of firm-years with Compustat to obtain firm controls and with SDC to identify mergers. As before, we match acquiring firms with nonacquiring control firms based on their propensity-to-acquire score generated from the Harford (1999) model.

Using this larger sample and extending the time period to 10 years before and after the merger, Figure 8 shows innovation differences between acquiring and nonacquiring firms using a model adapted from equation (4), and Panel C of Table IX presents regression results analogous to those in Panel A. Even over this longer time period, we fail to identify significant upticks in innovation within acquiring firms relative to nonacquiring firms. In fact, we no longer observe jumps in secondary applications or labeling and manufacturing-related applications.

Taken as a whole, our innovation results suggest that any increase in innovation around mergers is a recent phenomenon concentrated within follow-on applications to relabel products or streamline manufacturing. Although this sort of innovation likely benefits shareholders by increasing sales or cost efficiencies, it is unlikely to serve consumers.

VI. Conclusion

Prior studies document that M&As tend to generate wealth for shareholders. We confirm this finding in the context of recent mergers in the pharmaceutical industry and explore whether shareholder value creation tends to stem from synergies or market power gains on net. We find that pharmaceutical mergers are generally accompanied by increases in product prices—particularly within uncompetitive product markets that experience further consolidation as a result of the merger. We also examine innovation around mergers and find that innovation activity is limited to labeling and manufacturing process changes, not new drug creation. These findings are inconsistent with synergistic gains being passed along to consumers through lower prices or better products.

Our study has implications for policymakers, who often claim that ensuring affordable access to medication for constituents is a top priority. We show that one contributor to rising drug prices is recent consolidation in the pharmaceutical industry. Understanding the nature of competitive forces in this industry provides insights into how to better regulate this important industry and contain drug prices.

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Appendix

Table AI
Acquisition Likelihood Model Variable Definitions

This table presents variable definitions of our variables used in the Harford (1999) acquisition likelihood model. Variables are constructed from CRSP and Compustat data.

Harford (1999) variables	
Sales growth	Percentage growth in sales, averaged from year $t-4$ to $t-1$.
Noncash working capital	Current assets minus current liabilities and cash and cash equivalents, divided by total assets, averaged from year $t-4$ to $t-1$.
Leverage	Book value of debt divided by the market value of equity, averaged from year $t-4$ to $t-1$.
Market-to-book	The market value of equity divided by the book value of equity, averaged from year $t-4$ to $t-1$.
Price-to-earnings	Stock price divided by earning per share, averaged from year $t-4$ to $t-1$.
Size	The natural log of total assets at the beginning of year t.
Cash deviation	The difference between the firm's cash and cash equivalents as a percentage of total assets and the predicted industry average cash ratio at the beginning of year t.
Avg. abnormal returns	The average daily abnormal percentage returns estimated from a market model using daily returns over the prior year.

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Appendix S1: Internet Appendix.
Replication Code.